

LAB 3 – ARCHITECTURE SELECTION JUSTIFICATION

Digital Campus Library Reservation Gateway

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Architecture Selection: We chose Layered Architecture for the Digital Campus Library Reservation Gateway System.

Architectural Choice: We selected Layered Architecture as the architectural style for the system. It separates the presentation or user-interface layer, application and business-logic responsibilities, and the data-management layer. This provides a clear structure for the major library functions and their interactions.

Reason 1 – Clear separation of responsibilities: The library gateway contains several distinct functions, including ISBN catalog search, book reservation, FIFO holding-queue management, availability notifications, and overdue fine management. Layered Architecture allows these responsibilities to be organized into separate components while keeping persistent information in the database layer. This makes the system easier to understand, develop, test, and maintain.

Reason 2 – Support for Different User Roles:The system has two main stakeholders, Student Members and the Head Librarian, who perform different operations. Students can search books, place holds, view queues, and receive notifications, while the librarian can manage reservations and overdue fines. A layered structure provides a clear path from user requests to the appropriate business component and then to the required library data.

Security Advantage: Security can be strengthened by applying authentication and role-based authorization in the interface and application layers. Student Members can be restricted to student functions, while only authorized Head Librarians can perform librarian operations such as managing reservations and fines. This helps prevent unauthorized access to role-specific functions.

Performance Benefit: Separating the catalog-search component from the database layer allows the system to perform focused and efficient database queries instead of processing unnecessary information. This supports paginated ISBN catalog searches and helps the system meet the performance requirement of returning results for up to 50,000 titles in less than 150 ms.

Component Diagram

